GRAPH CONVOLUTION NUERAL NETWORK

A Brief Introduction · XU Chentianye

OUTLINE

This presentation is divided into four sections:

1. Introduction
2. Establishment of Graph Convolution Neural Network Layer
3. Application (including a comparison test)
4. Conclusion and Discussion

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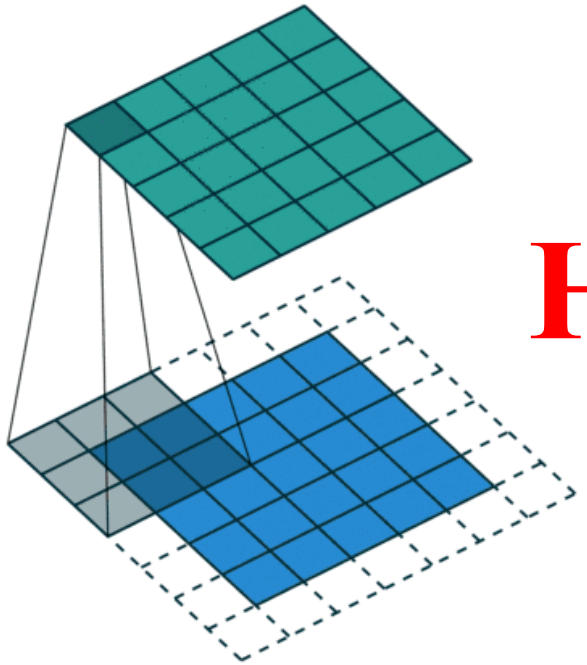
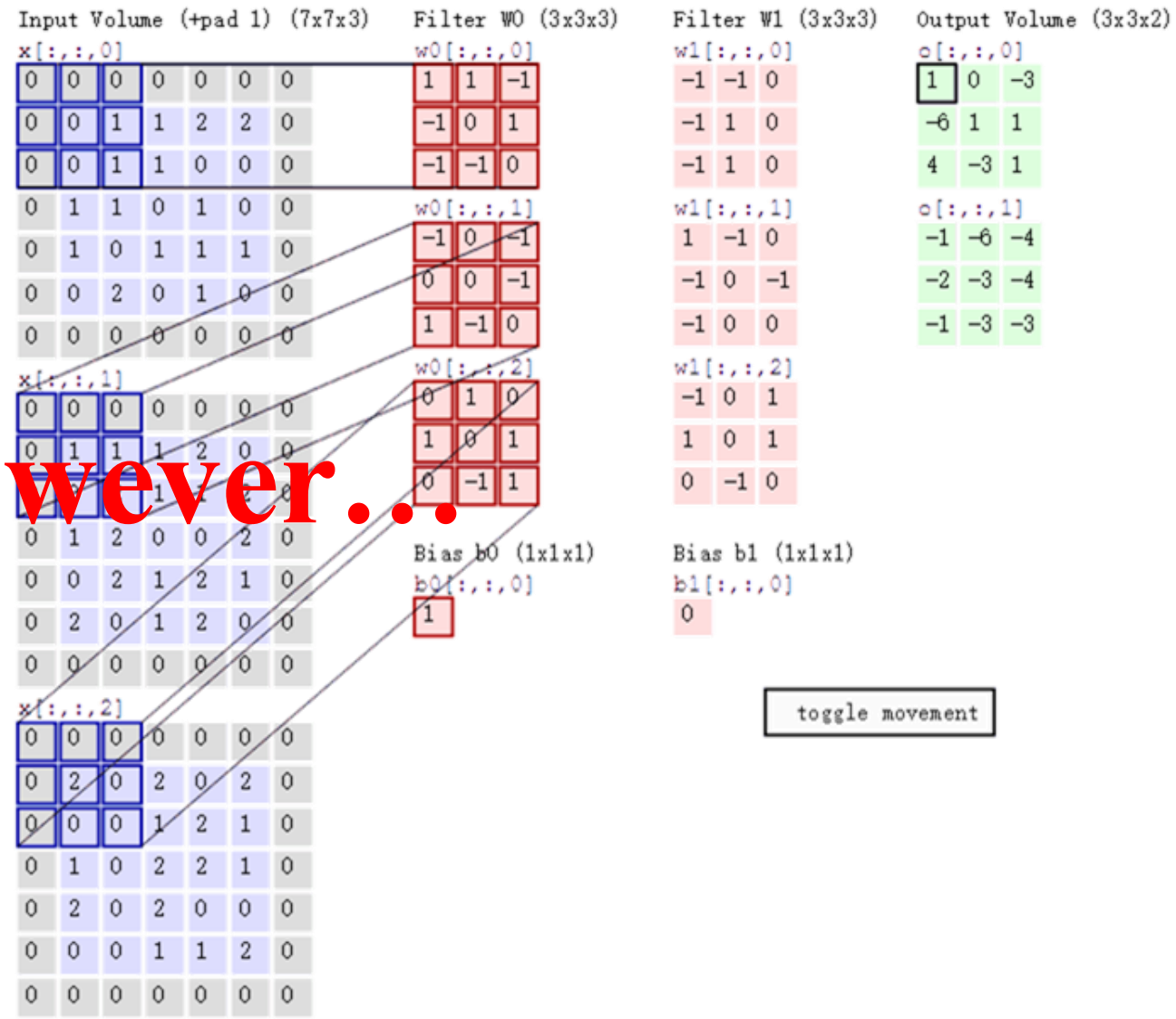
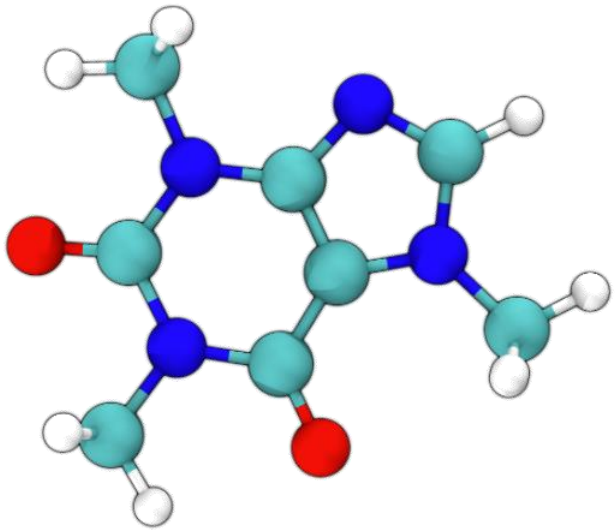
A BRIEF LOOK BACK: Convolution Neural Network

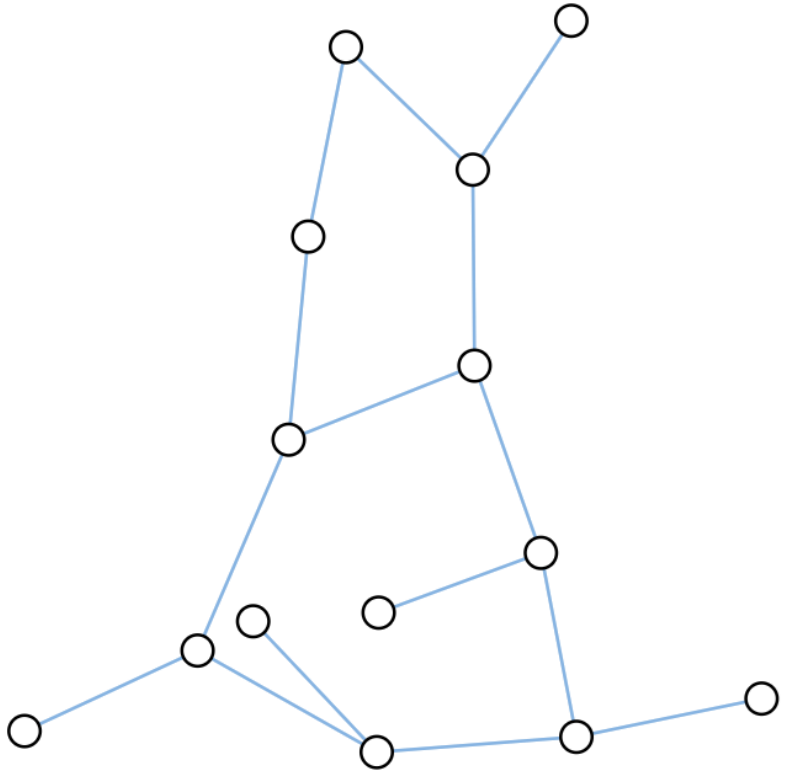


However…



A THORNY CASE





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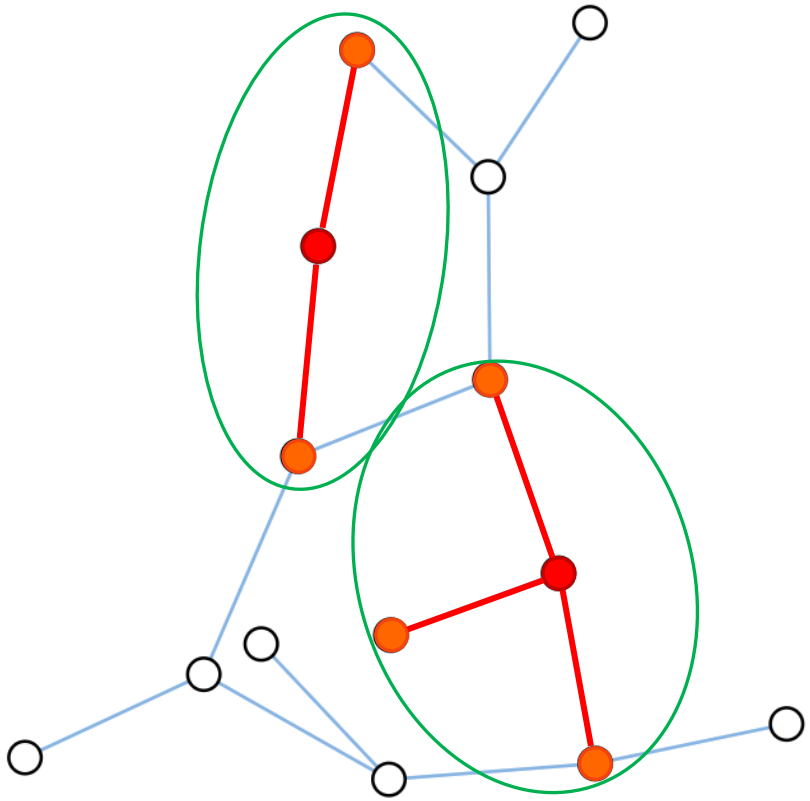
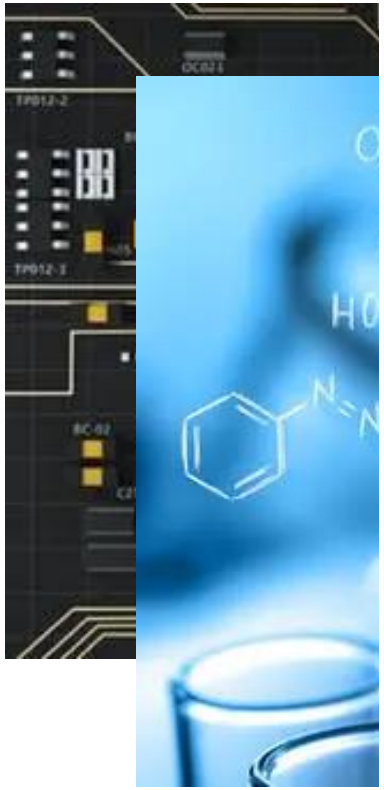
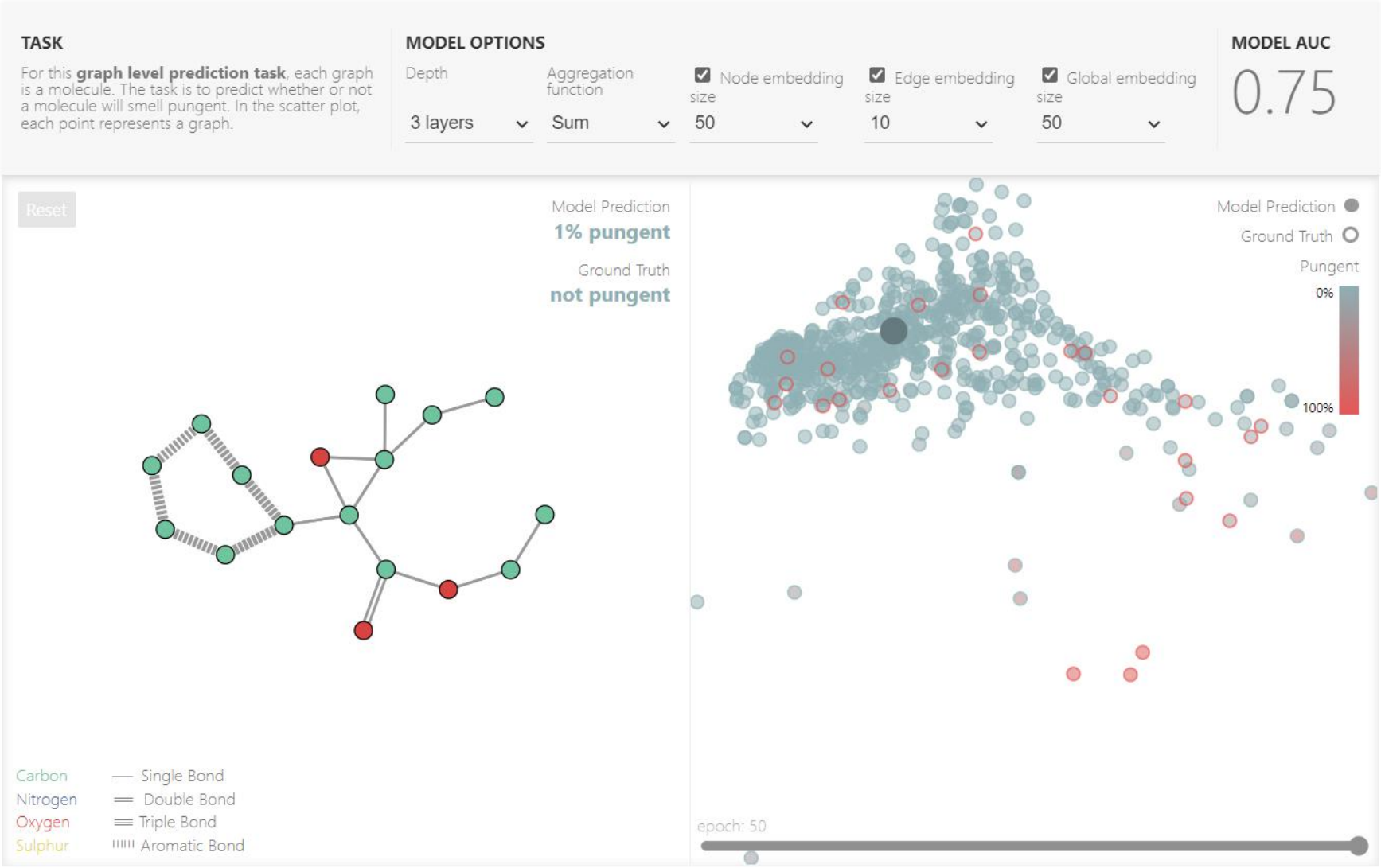


Nodes with different neighbor numbers!

Fixed-size filters are not applicable any more.

A BOOMING INDUSTRY…





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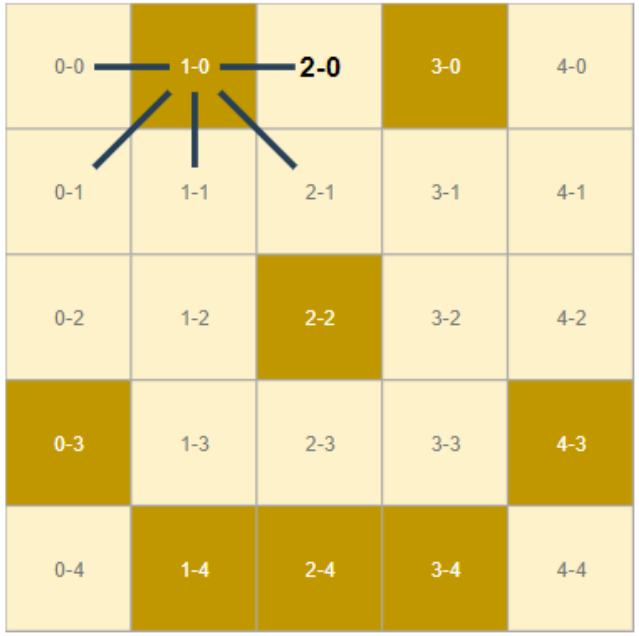
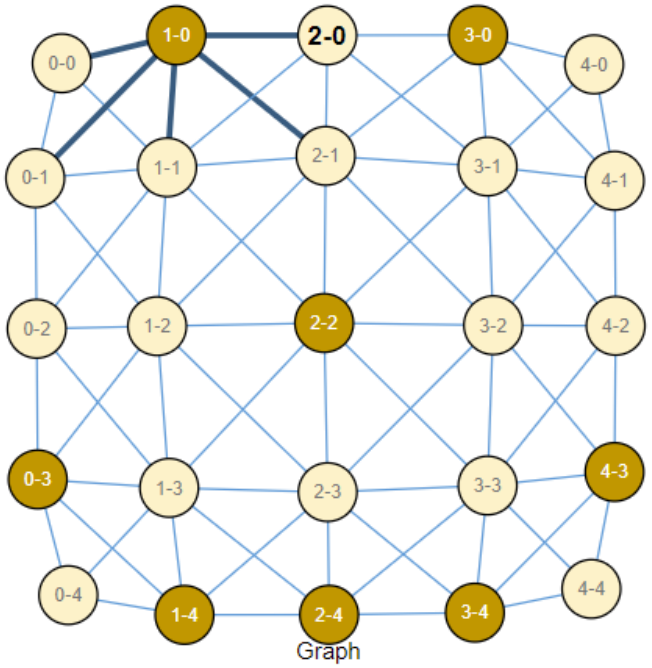
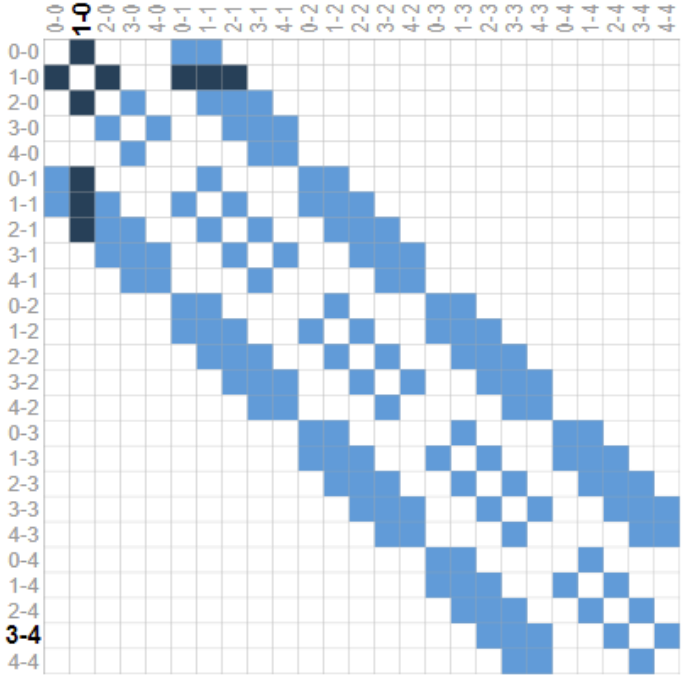
WHAT ARE GRAPHS?

View images graphically:

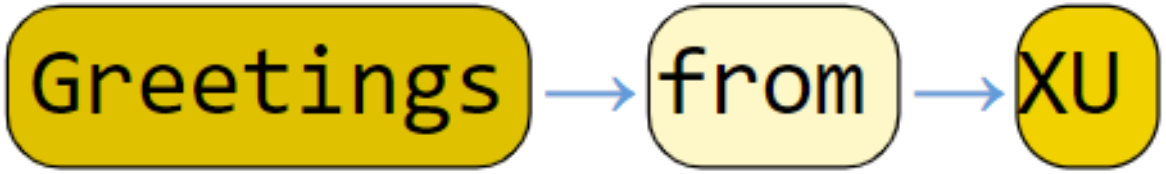


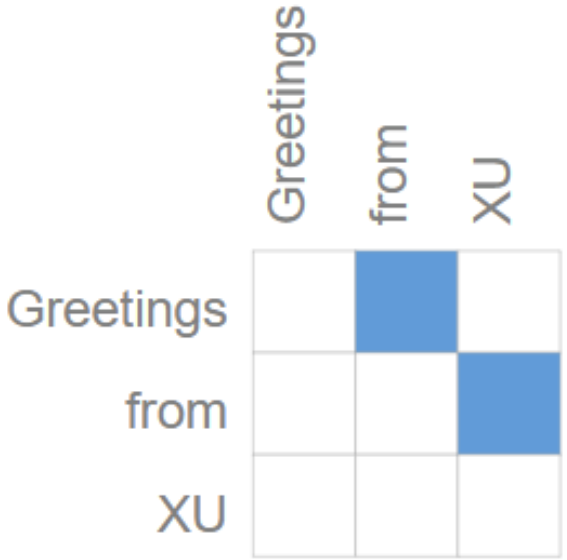




WHAT ARE GRAPHS?

View text graphically:





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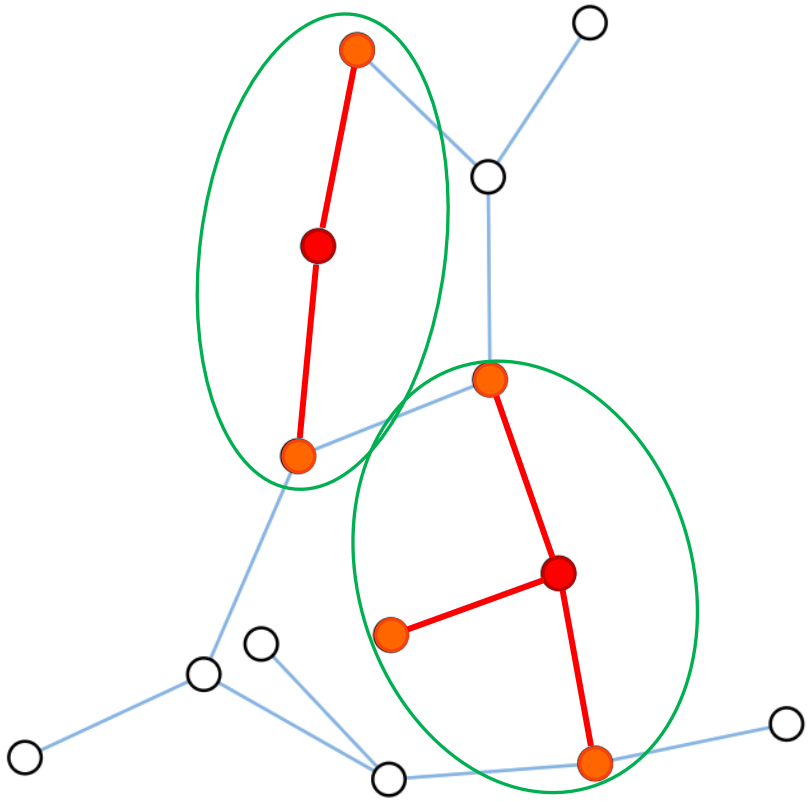
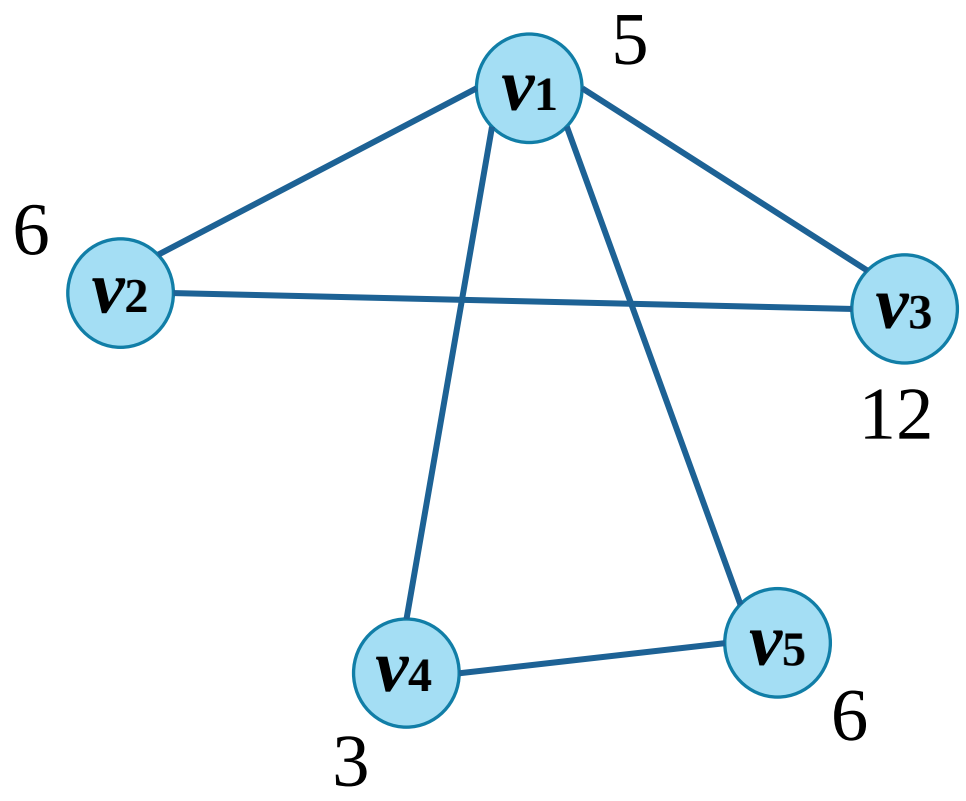


Nodes with different neighbor numbers!

Fixed-size filters are not applicable any more.

DEFINITION: Graph Signal



$$\boldsymbol{x} = [x_1, x_2, \cdots, x_N]^T$$

$$\boldsymbol{x} = [5, 6, 12, 3, 6]^T$$

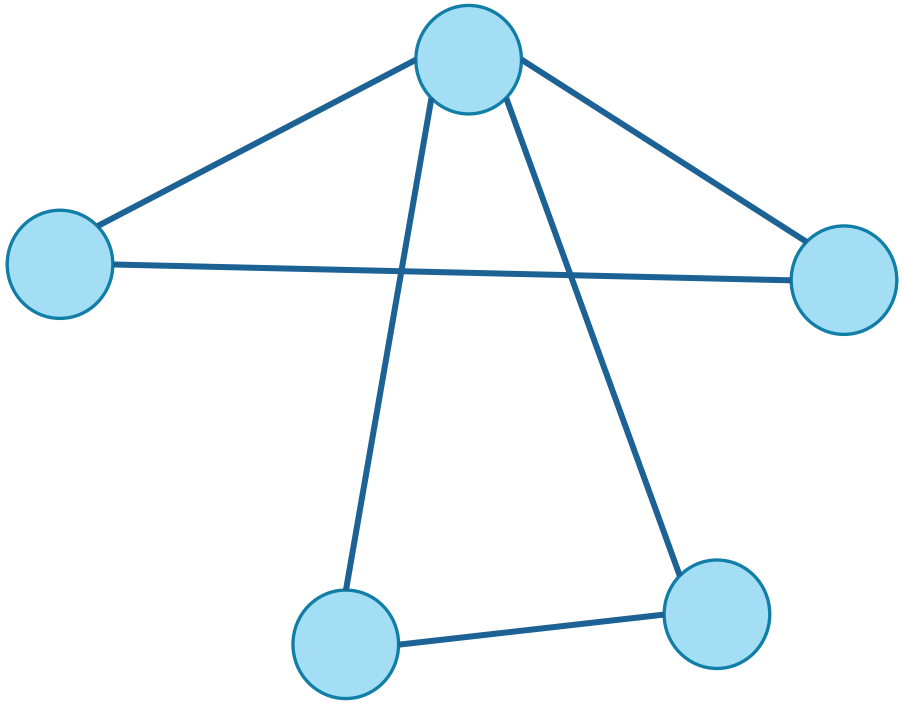
DEFINITION: Laplacian Matrix

$$\Delta f = \sum_{i=1}^{n} \frac{\partial^2 f}{\partial x_i^2} \longrightarrow$$

$$\begin{aligned}
\Delta f(x,y) &= \frac{\partial^2 f(x,y)}{\partial x^2} + \frac{\partial^2 f(x,y)}{\partial y^2} \\
&= [(f(x+1,y) - f(x,y)) - (f(x,y) - f(x-1,y))] + \\
&\quad [(f(x,y+1) - f(x,y)) - (f(x,y) - f(x,y-1))] \\
&= [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)] - 4f(x,y)
\end{aligned}$$

$$\longrightarrow \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

OUR GOAL…



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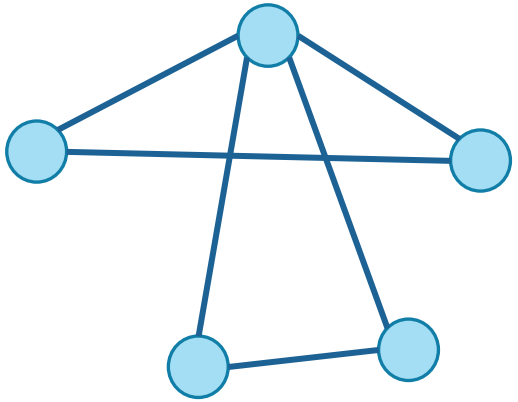

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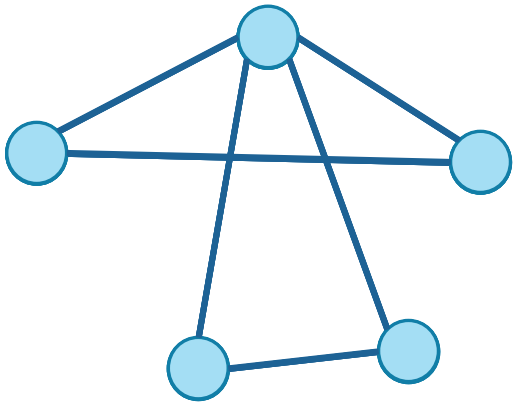

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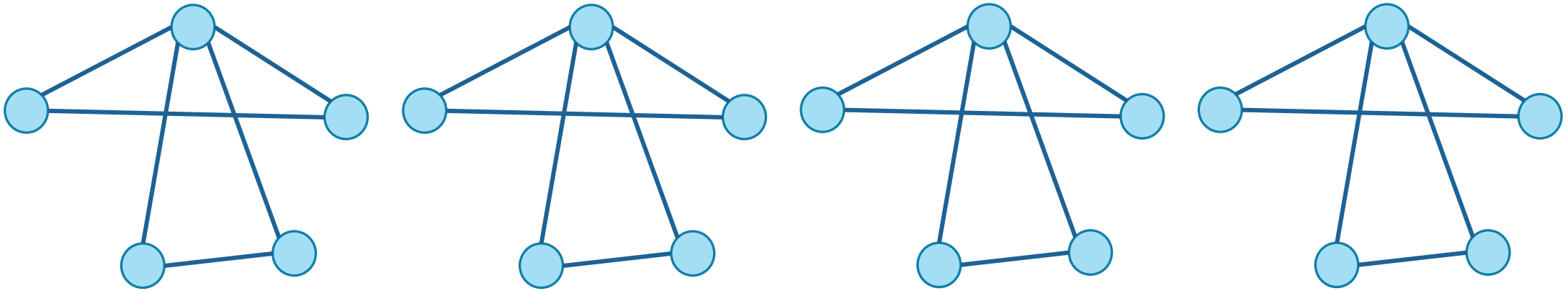

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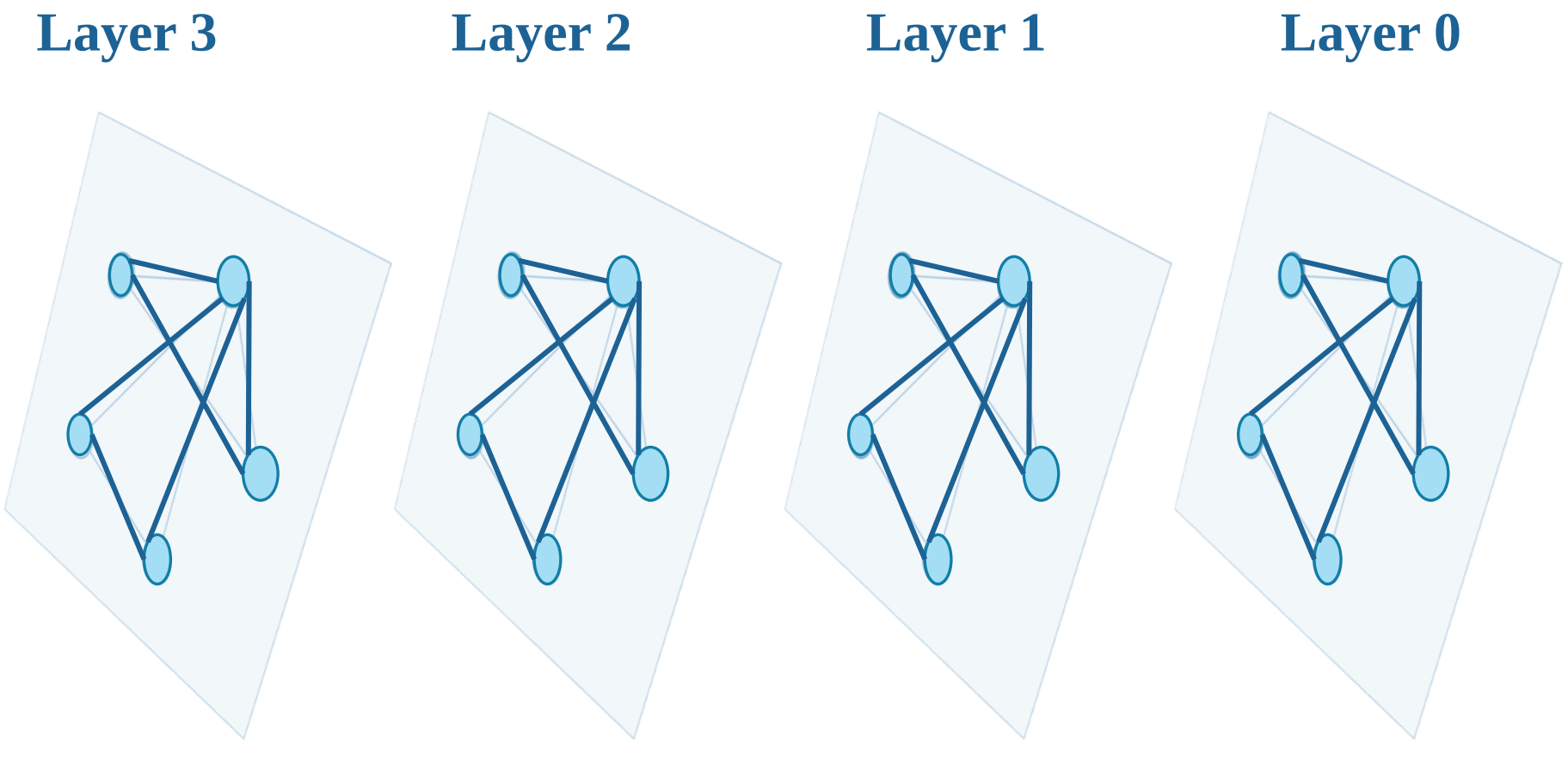

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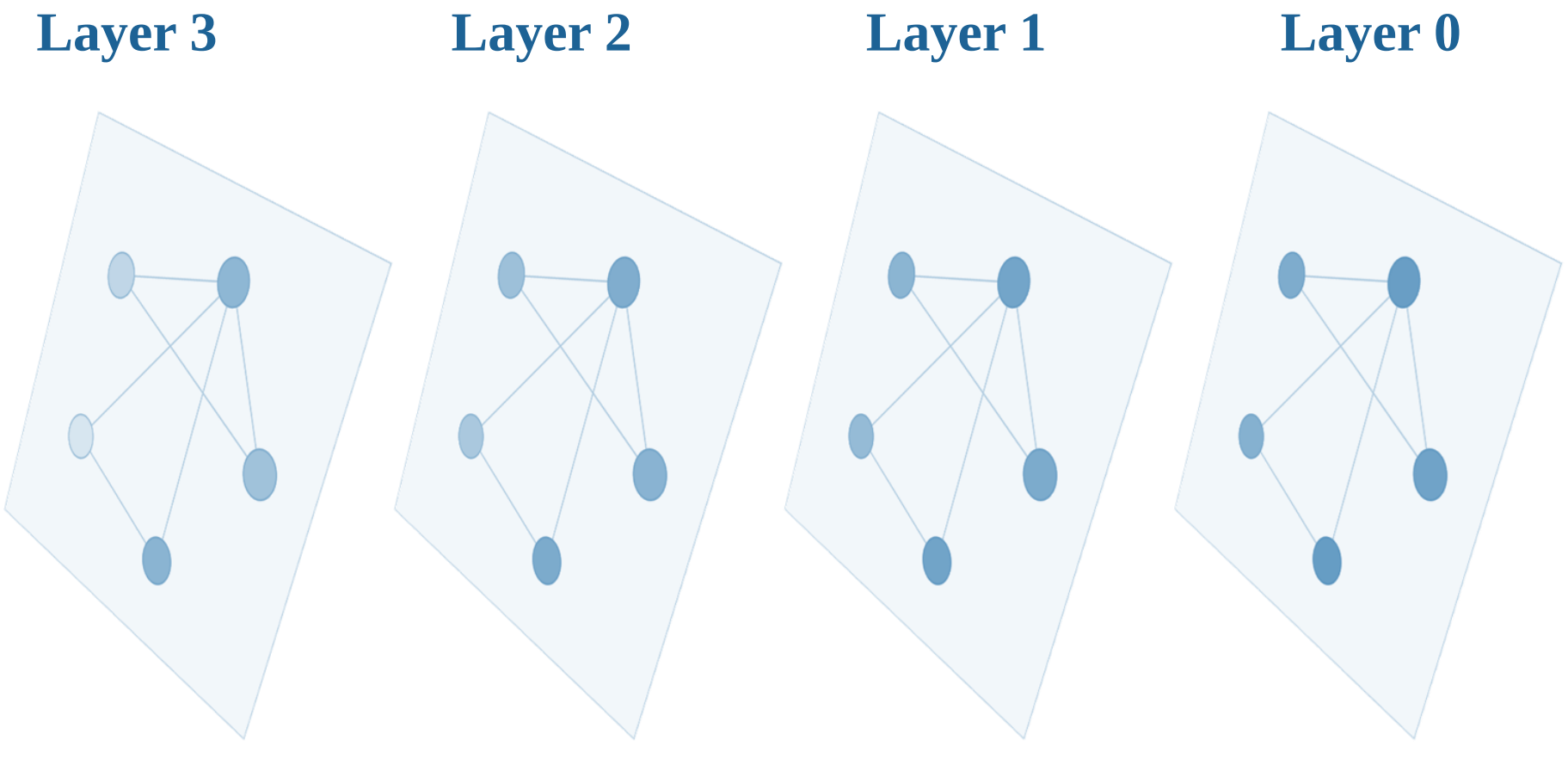

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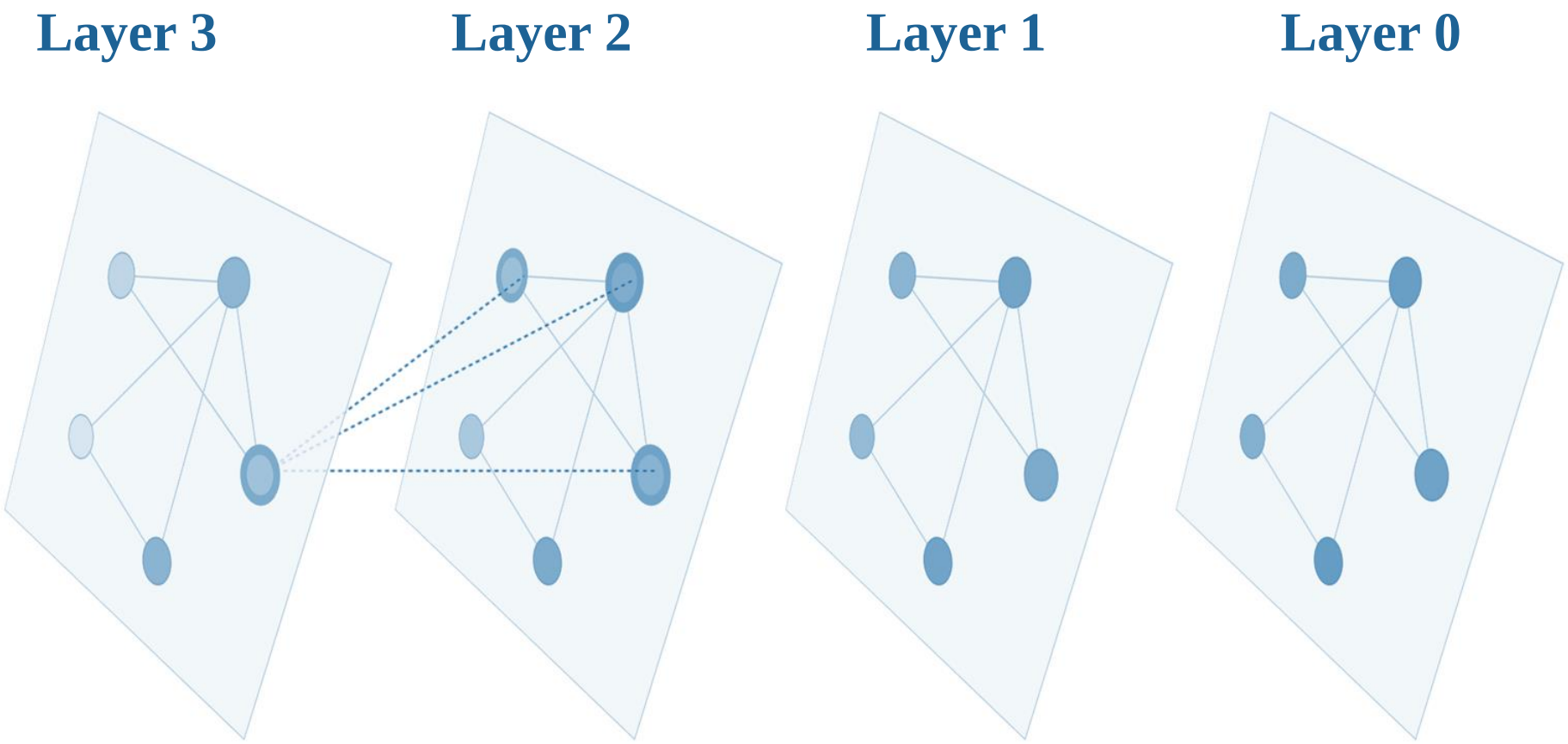

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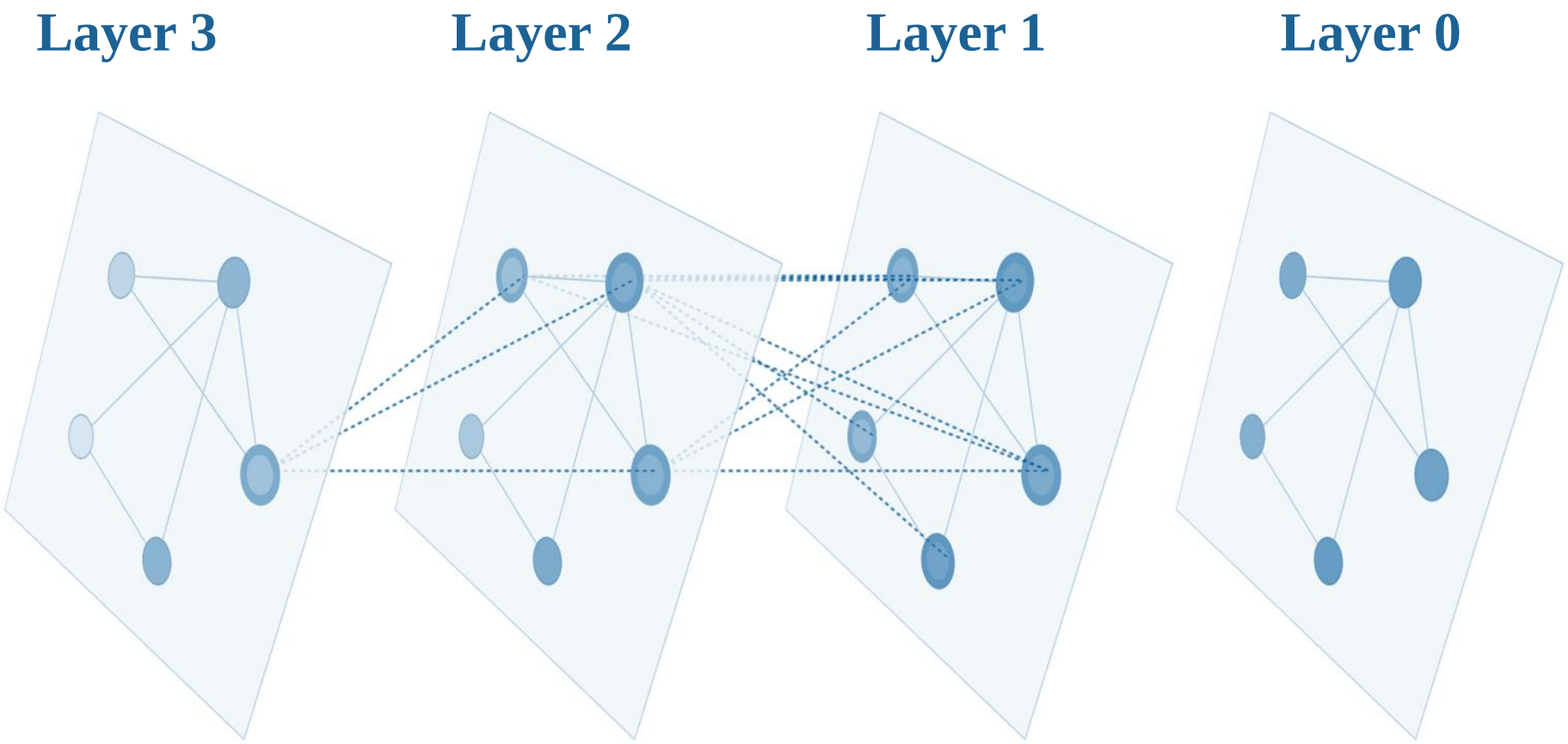

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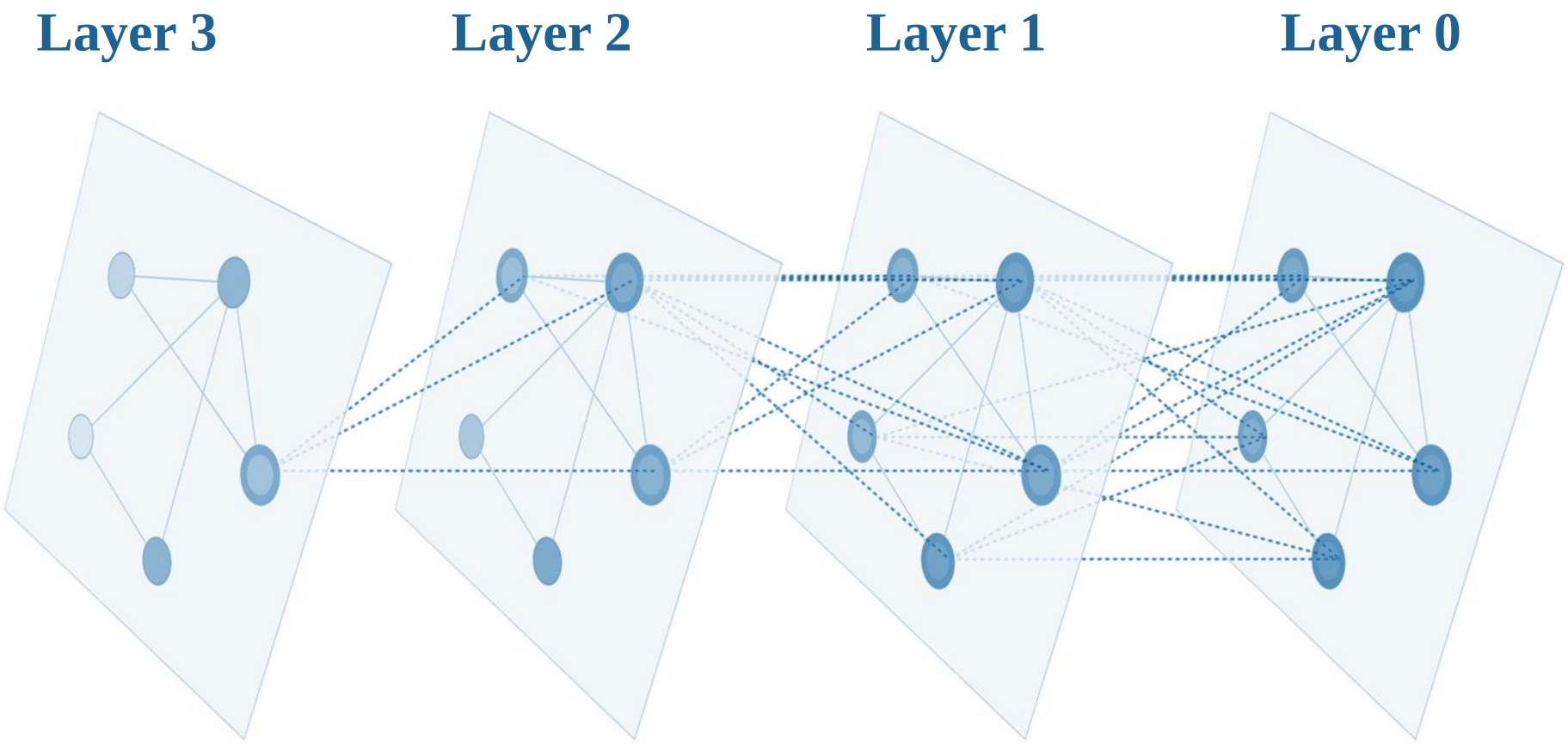

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$\boldsymbol{y} = L\boldsymbol{x} = [y_1, y_2, \cdots, y_N]^T$$

$$y_i = x_i \cdot \mathrm{Deg}(x_i) - \sum_{v_j \in N(v_i)} x_j$$

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} = \begin{bmatrix} x_1 \cdot \mathrm{Deg}(x_1) - \sum_{v_j \in N(v_1)} x_j \\ x_2 \cdot \mathrm{Deg}(x_2) - \sum_{v_j \in N(v_2)} x_j \\ \vdots \\ x_N \cdot \mathrm{Deg}(x_N) - \sum_{v_j \in N(v_N)} x_j \end{bmatrix} = \begin{bmatrix} x_1 \cdot \mathrm{Deg}(x_1) \\ x_2 \cdot \mathrm{Deg}(x_2) \\ \vdots \\ x_N \cdot \mathrm{Deg}(x_N) \end{bmatrix} - \begin{bmatrix} \sum_{v_j \in N(v_1)} x_j \\ \sum_{v_j \in N(v_2)} x_j \\ \vdots \\ \sum_{v_j \in N(v_N)} x_j \end{bmatrix}$$

$$= \begin{bmatrix} \mathrm{Deg}(x_1) & & & \\ & \mathrm{Deg}(x_2) & & \\ & & \cdots & \\ & & & \mathrm{Deg}(x_N) \end{bmatrix} \cdot \boldsymbol{x} - \boxed{\begin{bmatrix} \sum_{v_j \in N(v_1)} x_j \\ \sum_{v_j \in N(v_2)} x_j \\ \vdots \\ \sum_{v_j \in N(v_N)} x_j \end{bmatrix}}$$

$$A_{ij} = \begin{cases} 1, \text{ if } (v_i, v_j) \in E \\ 0, \text{ else} \end{cases}$$

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$\begin{bmatrix} \sum_{v_j \in N(v_1)} x_j \\ \sum_{v_j \in N(v_2)} x_j \\ \vdots \\ \sum_{v_j \in N(v_N)} x_j \end{bmatrix} = A\boldsymbol{x}$$

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$D = \begin{bmatrix} \mathrm{Deg}(x_1) & & & \\ & \mathrm{Deg}(x_2) & & \\ & & \cdots & \\ & & & \mathrm{Deg}(x_N) \end{bmatrix}$$

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$\boldsymbol{y} = \begin{bmatrix} \mathrm{Deg}(x_1) & & & \\ & \mathrm{Deg}(x_2) & & \\ & & \ddots & \\ & & & \mathrm{Deg}(x_N) \end{bmatrix} \cdot \boldsymbol{x} - \begin{bmatrix} \sum_{v_j \in N(v_1)} x_j \\ \sum_{v_j \in N(v_2)} x_j \\ \vdots \\ \sum_{v_j \in N(v_N)} x_j \end{bmatrix} = D\boldsymbol{x} - A\boldsymbol{x}$$

$$\boldsymbol{y} = L\boldsymbol{x} = [y_1, y_2, \cdots, y_N]^T$$

$$L = D - A$$

HOW TO AGGREGATE NEIGHBORS' FEATURES?

$$L\boldsymbol{x} = (D - A)\boldsymbol{x}$$

DEFINITION: Graph Fourier Transform & Inverse Graph Fourier Transform

$$L = V \Lambda V^T = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_N \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_N \end{bmatrix} \begin{bmatrix} \boldsymbol{v}_1^T \\ \boldsymbol{v}_2^T \\ \vdots \\ \boldsymbol{v}_N^T \end{bmatrix}$$

DEFINITION: Graph Fourier Transform &
Inverse Graph Fourier Transform

$$\boldsymbol{x}_{GFT_k} = \langle \boldsymbol{v}_k, \boldsymbol{x} \rangle$$

$$\boldsymbol{x}_{GFT} = V^T \boldsymbol{x}$$

$$V\boldsymbol{x}_{GFT} = VV^T \boldsymbol{x} = I\boldsymbol{x} = \boldsymbol{x}$$

$$\boldsymbol{x} = V\boldsymbol{x}_{GFT}$$

DEFINITION: Graph Fourier Transform & Inverse Graph Fourier Transform

$$
\begin{aligned}
\boldsymbol{x} &= V\boldsymbol{x}_{GFT} \\
&= \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_N \end{bmatrix} \begin{bmatrix} x_{GFT_1} \\ x_{GFT_2} \\ \vdots \\ x_{GFT_N} \end{bmatrix} \\
&= x_{GFT_1}\boldsymbol{v}_1 + x_{GFT_2}\boldsymbol{v}_2 + \cdots + x_{GFT_N}\boldsymbol{v}_N
\end{aligned}
$$

DEFINITION: Graph Filter

$$\boldsymbol{y} = H\boldsymbol{x} = \sum_{k=1}^{N} (h(\lambda_k) x_{GFT_k}) \boldsymbol{v}_k$$

DEFINITION: Graph Filter

$$\boldsymbol{y} = H\boldsymbol{x} = \sum_{k=1}^{N} (h(\lambda_k) x_{GFT_k}) \boldsymbol{v}_k = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_N \end{bmatrix} \begin{bmatrix} h(\lambda_1) x_{GFT_1} \\ h(\lambda_2) x_{GFT_2} \\ \vdots \\ h(\lambda_N) x_{GFT_N} \end{bmatrix}$$

$$= V \begin{bmatrix} h(\lambda_1) & & & \\ & h(\lambda_2) & & \\ & & \ddots & \\ & & & h(\lambda_N) \end{bmatrix} \begin{bmatrix} x_{GFT_1} \\ x_{GFT_2} \\ \vdots \\ x_{GFT_N} \end{bmatrix}$$

$$= V \boxed{\begin{bmatrix} h(\lambda_1) & & & \\ & h(\lambda_2) & & \\ & & \ddots & \\ & & & h(\lambda_N) \end{bmatrix}} V^T \boldsymbol{x}$$

$$\Lambda_h$$

DEFINITION: Graph Filter

$$H = V \begin{bmatrix} h(\lambda_1) & & & \\ & h(\lambda_2) & & \\ & & \ddots & \\ & & & h(\lambda_N) \end{bmatrix} V^T = V \Lambda_h V^T$$

FINALLY…

We can define graph convolution neural network now.

$$\boldsymbol{x}_1 * \boldsymbol{x}_2 = \mathrm{IGFT}(\mathrm{GFT}(\boldsymbol{x}_1) \odot \mathrm{GFT}(\boldsymbol{x}_2))$$

$$\begin{aligned}
\boldsymbol{x}_1 * \boldsymbol{x}_2 &= V((V^T \boldsymbol{x}_1) \odot (V^T \boldsymbol{x}_2)) \\
&= V(\boldsymbol{x}_{GFT_1} \odot (V^T \boldsymbol{x}_2)) \\
&= V(\mathrm{diag}(\boldsymbol{x}_{GFT_1})(V^T \boldsymbol{x}_2)) \\
&= \boxed{(V\,\mathrm{diag}(\boldsymbol{x}_{GFT_1})V^T}\,)\boldsymbol{x}_2
\end{aligned}$$

FINALLY…

We can define graph convolution neural network now.

(1) Parameterize Frequency Response Matrix:

$$X' = \sigma\left(V \begin{bmatrix} \theta_1 & & & \\ & \theta_2 & & \\ & & \ddots & \\ & & & \theta_N \end{bmatrix} V^T X \right) = \sigma(\Theta X)$$

FINALLY…

We can define graph convolution neural network now.

(2) Polynomial Fit of the Frequency Response Matrix:

$$H = h_0 L^0 + h_1 L^1 + h_2 L^2 + \cdots + h_K L^K = \sum_{k=0}^{K} h_k L^k$$

$$X' = \sigma\left(V \left(\sum_{k=0}^{K} \theta_k \Lambda^k \right) V^T X \right) = \sigma\left(V \operatorname{diag}(\psi\theta) V^T X \right)$$

FINALLY…

We can define graph convolution neural network now.

(3) Fixed Graph Filter:

$$X' = \sigma\left(\theta(I + L)X\right) = \sigma\left(\theta L' X\right)$$

$$X' = \sigma\left(L' XW\right)$$

FINALLY…

We can define graph convolution neural network now.

(3) Fixed Graph Filter:

STEP 2 图卷积层的定义

```
class GraphConvolution(nn.Module):
    def __init__(self, inDim, outDim):
        super(GraphConvolution, self).__init__()
        self.inDim = inDim
        self.outDim = outDim
        self.weight = nn.Parameter(torch.Tensor(inDim, outDim))
        init.kaiming_uniform_(self.weight)
        self.bias = nn.Parameter(torch.Tensor(outDim))
        init.zeros_(self.bias)

    def forward(self, A, inp):
        return torch.sparse.mm(A, torch.mm(inp, self.weight)) + self.bias
```

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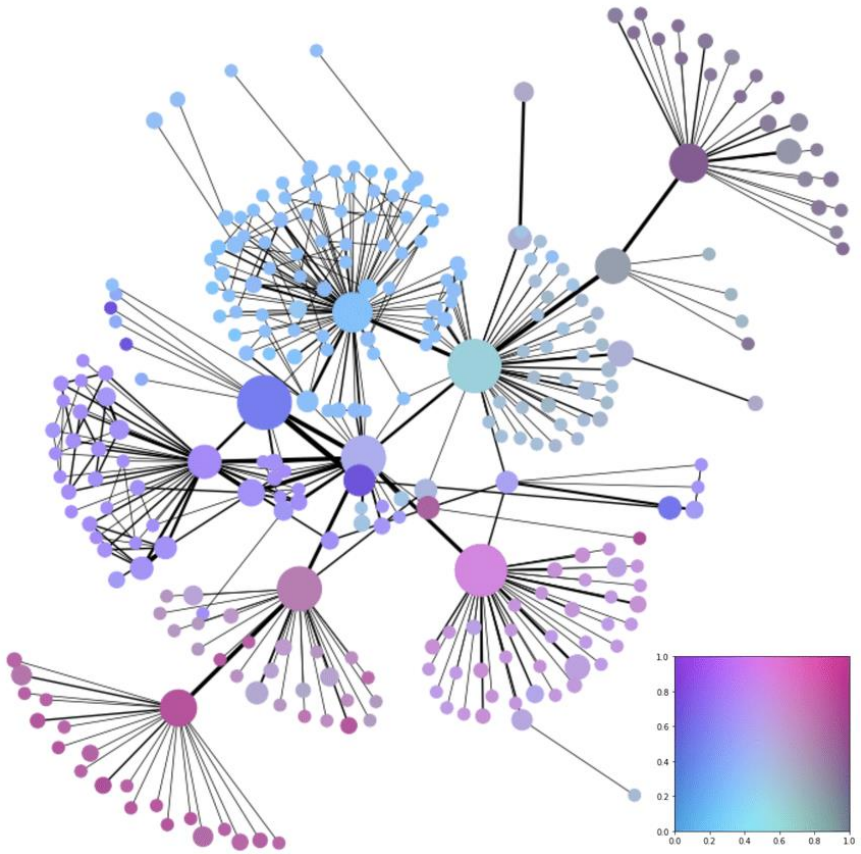
ABOUT THE DATASET

DGM Visualisation of the Cora Dataset



NETWORK DESIGN

Graph Convolution Neural Network

STEP 3 图卷积神经网络的定义

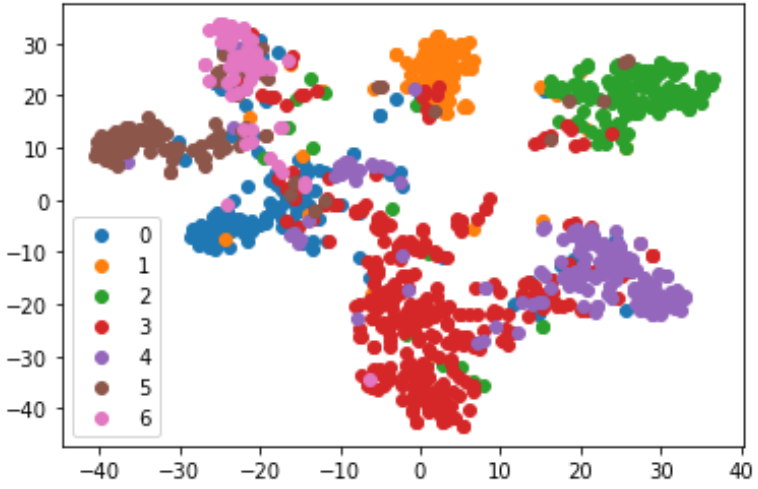
```
class GCN(nn.Module):
    def __init__(self):
        super(GCN, self).__init__()
        self.gcn1 = GraphConvolution(1433, 500)
        self.gcn2 = GraphConvolution(500, 100)
        self.gcn3 = GraphConvolution(100, 7)

    def forward(self, A, inp):
        outp1 = F.relu(self.gcn1(A, inp))
        outp2 = F.relu(self.gcn2(A, outp1))
        outp3 = self.gcn3(A, outp2)
        return outp3
```

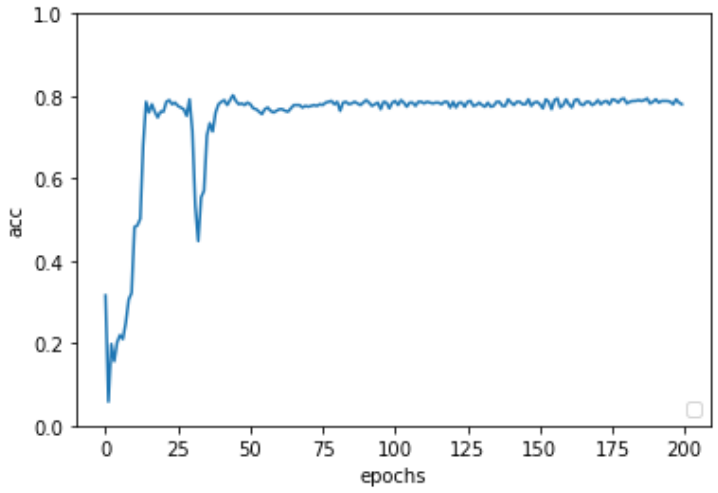
NETWORK DESIGN

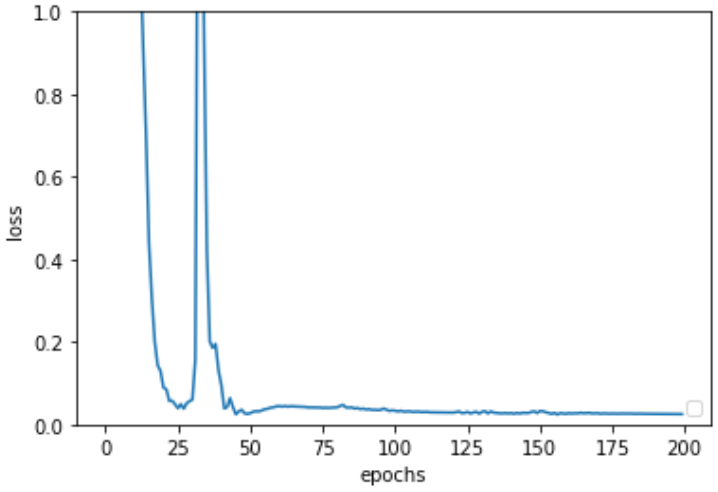
How about the Fully Connected Neural Network?

EXPERIMENT RESULT & ANALYSIS

```
Epoch 187: Loss 0.0277, TrainAcc 1.0000, ValAcc 0.8080
Epoch 188: Loss 0.0267, TrainAcc 1.0000, ValAcc 0.8060
Epoch 189: Loss 0.0275, TrainAcc 1.0000, ValAcc 0.8080
Epoch 190: Loss 0.0266, TrainAcc 1.0000, ValAcc 0.8060
Epoch 191: Loss 0.0273, TrainAcc 1.0000, ValAcc 0.8100
Epoch 192: Loss 0.0266, TrainAcc 1.0000, ValAcc 0.8080
Epoch 193: Loss 0.0269, TrainAcc 1.0000, ValAcc 0.8040
Epoch 194: Loss 0.0273, TrainAcc 1.0000, ValAcc 0.8120
Epoch 195: Loss 0.0266, TrainAcc 1.0000, ValAcc 0.8100
Epoch 196: Loss 0.0266, TrainAcc 1.0000, ValAcc 0.8060
Epoch 197: Loss 0.0270, TrainAcc 1.0000, ValAcc 0.8120
Epoch 198: Loss 0.0266, TrainAcc 1.0000, ValAcc 0.8080
Epoch 199: Loss 0.0264, TrainAcc 1.0000, ValAcc 0.8080
Test acc:  0.8100000023841858
```

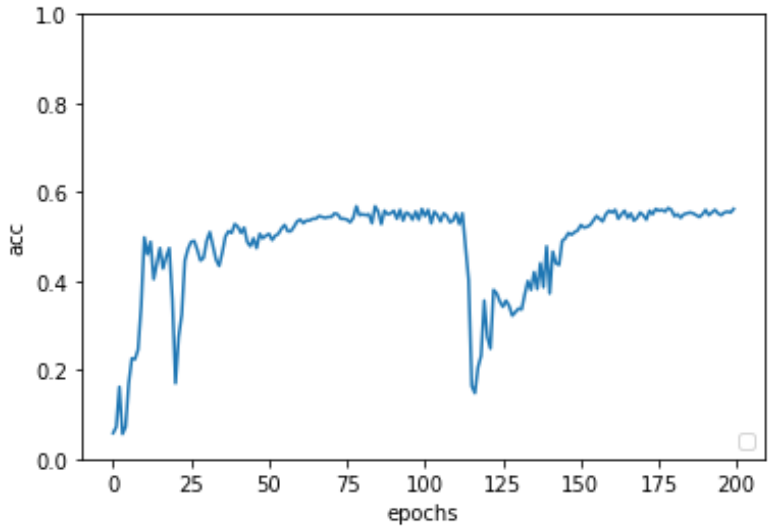


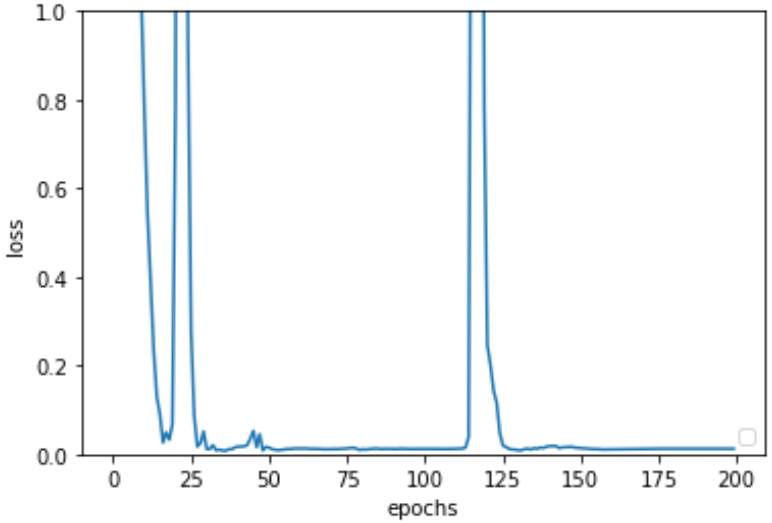




EXPERIMENT RESULT & ANALYSIS

```
Epoch 190: Loss 0.0116, TrainAcc 1.0000, ValAcc 0.4920
Epoch 191: Loss 0.0114, TrainAcc 1.0000, ValAcc 0.5040
Epoch 192: Loss 0.0115, TrainAcc 1.0000, ValAcc 0.4940
Epoch 193: Loss 0.0115, TrainAcc 1.0000, ValAcc 0.4960
Epoch 194: Loss 0.0114, TrainAcc 1.0000, ValAcc 0.4940
Epoch 195: Loss 0.0115, TrainAcc 1.0000, ValAcc 0.4980
Epoch 196: Loss 0.0114, TrainAcc 1.0000, ValAcc 0.4960
Epoch 197: Loss 0.0114, TrainAcc 1.0000, ValAcc 0.4860
Epoch 198: Loss 0.0115, TrainAcc 1.0000, ValAcc 0.4920
Epoch 199: Loss 0.0114, TrainAcc 1.0000, ValAcc 0.4960
Test acc:  0.4959999918937683
```





OUTLINE

This presentation is divided into four sections:

1. Introduction
2. Establishment of Graph Convolution Neural Network Layer
3. Application (including a comparison test)
4. Conclusion and Discussion ☜

CONCLUSION & DISCUSSION

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THANK YOU!

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